Autothrottle, FADEC & Autopilot Operating Manual — Cirrus Vision SF50 G2 (X-Plane 12)

This operational guide provides real-world and simulated operating procedures for the **Perspective Touch+ digital Autothrottle / FADEC system** and the integrated Garmin GFC 700 digital autopilot on the **Cirrus Vision SF50 G2** in X-Plane 12.

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1. Overview of Touchscreen Controls (Center GTC)

On the primary Autothrottle touchscreen page on the Garmin Touchscreen Controller (GTC), there are **8 interactive touch buttons** and **1 dual-concentric rotary knob (`SPEED SET`)**:

Left Column (Speed Management & Master Autothrottle)

- **`Auto T` (`ON` / `OFF`)**: Master Autothrottle Switch.
- When **engaged (Green bar)**: The autothrottle actively servos the thrust lever to capture and maintain the selected speed target (`KTS` or `Mach`), while automatically enforcing the continuous rating limit (**`MCT`**).
- When **pressed again (Off / Gray bar)**: Instantly disconnects the entire autothrottle system (`Auto T`, `MCT`, `TOGA`, and `FMS` links), returning **100% manual throttle authority** to your physical hardware throttle lever with no residual servo bias.
- **`KTS`**: Selects Indicated Airspeed hold mode in **Knots (`IAS`, range `85 - 250 KIAS`)**. If switching from `Mach` mode, it automatically converts the current flight Mach number into the equivalent Indicated Airspeed.
- **`Mach`**: Selects speed hold mode in **Mach Number (`0.250 - 0.535 M`)**. If switching from `KTS` mode, it seamlessly calculates and displays the equivalent Mach number at the current pressure altitude and outside air temperature.
- **`FMS`**: Toggles coupling to the FMS automated speed schedule (coupling into the `M CRS` / `E CRS` cruise profiles). Pressing it while active transitions back to manual speed selection (`MAN`) while maintaining the current target speed.

- **`SPEED SET` Rotary Knob (Bottom-right corner of the touchscreen bezel)**:
- In **`KTS` mode**: Adjusts target airspeed in increments of `1 kt` (inner knob) or `10 kts` (outer knob).
- In **`Mach` mode**: Adjusts target Mach in increments of `0.001 M` (inner knob) or `0.010 M` (outer knob).
- If the knob is rotated while an automated FMS preset (`M CRS` or `E CRS`) is active, the system automatically transitions from `FMS` mode to manual `Mach` mode at the newly dialed value.

Right Column (FADEC Thrust Ratings & G2 Cruise Presets)

- **`TOGA` (Takeoff / Go-Around)**:
- Smoothly commands the Williams International FJ33-5A turbofan to **100% Takeoff Rated Thrust (`TLA = 1.00`)**, activates the Flight Director to `+8°` pitch-up target attitude (if off), and pre-sets the climb airspeed bug to **`160 KIAS`**.
- **`MCT` (Maximum Continuous Thrust)**:
- Enforces the FADEC **Maximum Continuous Thrust limit (`95%` thrust / `TLA = 0.90`)** and activates the Autothrottle (**`Auto T ON`**):
    - If climbing in **`FLC` (Flight Level Change)** autopilot mode: Holds the throttle locked precisely on the **`MCT` detent** while autopilot pitch commands maintain the commanded climb airspeed.
    - In **level cruise (`ALT`)**, **`VS` (Vertical Speed)** climbs/descents, or manual flight: Linearly modulates engine thrust to maintain the selected target speed (`KTS` or `Mach`) without ever exceeding the continuous `MCT` threshold.
- **`M CRS` (Max Cruise — `0.530 Mach`)**:
- Automatically engages **`Mach` mode**, enables **`FMS`**, **`Auto T`**, and **`MCT`**, and locks the cruise target to **`0.530 Mach`** (**~311 KTAS** between `FL280` and `FL310`, fuel flow ~`65 gal/h`, typical range **950 NM**).
- **`E CRS` (Economy Cruise — `0.489 Mach`)**:
- Automatically engages **`Mach` mode**, enables **`FMS`**, **`Auto T`**, and **`MCT`**, and locks the cruise target to **`0.489 Mach`** (**~280 KTAS** at `FL310`, fuel flow ~`44 gal/h`, maximum range up to **1,275 NM**).

2. Standard Operating Procedures (SOP) by Flight Phase

Phase 1: Pre-Flight, Ground Handling & Taxi

1. **Engine Start & Ground Power (GPU)**:
2. On the **SYSTEMS** page of the center GTC, tap the dedicated **`GPU`** button to connect external ground power (green indicator bar illuminates). All avionics, environmental control, and cabin lights are powered at 28V without draining the main aircraft batteries.
3. *Safety Feature*: The GPU will automatically disconnect as soon as ground roll begins (`groundspeed > 3 kts`) or upon takeoff.
4. **Taxi**:
5. On the **SYSTEMS** page, confirm that **`NWS`** **(Nose Wheel Steering)** is active (**Green bar**) for smooth rudder-pedal directional control on the ground.
6. Extend **Flaps to 50%** ($V_{FE} = 190\text{ KIAS}$).
7. Set the initial cleared altitude using the **`ALT`** selector knob on the autopilot panel (calibrated to **`100 ft`** per click).
8. **Runway Line-Up**:
9. Once cleared for takeoff, tap **`TOGA`** on the touchscreen:
    - Both **`TOGA`** and **`Auto T`** illuminate green.
    - The airspeed target bug sets automatically to **`160 KIAS`**.
    - The Williams FJ33-5A engine spools up smoothly to **100% Takeoff Thrust**.

Phase 2: Takeoff & Initial Climb

1. **Rotation & Initial Climb**:
2. Smoothly rotate at **`90 KIAS`** toward the `8°–10°` pitch-up bars on the Primary Flight Display (PFD).
3. With a positive rate of climb established, retract the landing gear (**Gear UP**, limit $V_{LO} = 210\text{ KIAS}$).
4. Passing **`115 KIAS`**, retract the **Flaps to 0% (UP)**.

5. **Transition to `MCT` and Enroute Climb (`FLC` or `VS`)**:
6. After flap retraction (and within 5 minutes of takeoff), tap **`MCT`** on the touchscreen:
    - `TOGA` extinguishes; **`MCT`** and **`Auto T`** remain active in green.
7. **Recommended Climb Technique — `FLC` (Flight Level Change)**:
    1. Press **`FLC`** and **`AP`** on the G1000 autopilot control panel.
    2. With speed set to **`160 KIAS`** (up to ~FL200–FL220) and transitioning to **`Mach`** (`0.400 – 0.430 M`) climbing through FL250 towards **`FL280 – FL310`**, the autothrottle holds engine power fixed on the **`MCT` detent** while autopilot pitch control locks climb speed dead-center without oscillation.
8. **Alternative Climb Technique — `VS` (Vertical Speed)**:
    1. Press **`VS`** and dial the desired vertical speed (e.g., `+1,500 ft/min` at low altitude, decaying to `+600 / +800 ft/min` above FL250).
    2. In `VS` mode, the Autothrottle (`Auto T` + `MCT`) modulates the thrust lever continuously to maintain the dialed airspeed up to the continuous `MCT` limit.

Phase 3: Level-Off & Cruise (`FL280 – FL310`)

Upon reaching the target altitude (e.g., **`31,000 ft / FL310`**), the autopilot smoothly captures and transitions from `FLC`/`VS` to **`ALT` (Altitude Hold)**. Simultaneously, the autothrottle transitions automatically to **active speed hold**.

Select one of **3 Cruise Profiles**:

1. **Maximum Cruise Profile (`M CRS` — `0.530 Mach` / `311 KTAS`)**:
2. Tap **`M CRS`** on the touchscreen.
3. The status indicators **`M CRS`**, **`FMS`**, **`Mach`**, **`MCT`**, and **`Auto T`** illuminate in green.
4. The target display shows **`0.530 M`**. The autothrottle stabilizes between `0.528 – 0.530 Mach` (~`311 KTAS` at ISA, fuel burn ~`65 gal/h`, practical range **~950 NM**).
5. **Economy / Long-Range Cruise Profile (`E CRS` — `0.489 Mach` / ~`280 KTAS`)**:
6. Tap **`E CRS`** on the touchscreen.
7. The status indicators **`E CRS`**, **`FMS`**, **`Mach`**, **`MCT`**, and **`Auto T`** illuminate in green.
8. The target display shows **`0.489 M`**. If previously cruising faster (e.g., at `0.528 M`), the

autothrottle **smoothly retards the throttle** (to ~`68–72%` TLA), allows the airframe to decelerate cleanly to **`0.489 Mach`**, and then restabilizes thrust to hold **`0.489 Mach`** (fuel burn ~`44 gal/h`, maximum range up to **1,275 NM**).

9. **Manual Cruise Speed Selection (`Mach` or `KTS`)**:
10. Tap **`Mach`** (or **`KTS`**) and turn the **`SPEED SET`** knob to dial your custom target (e.g., `0.510 M` or `210 KIAS`).

Phase 4: Descent

1. Set the cleared lower altitude using the autopilot **`ALT`** knob (e.g., `10,000 ft` or pattern altitude).
2. **Descent in `VS` (Vertical Speed — Recommended)**:
3. Press **`VS`** and set `-1,000` to `-1,500 ft/min`.
4. Switch to **`KTS`** mode (when descending below FL240) and set the desired descent speed using **`SPEED SET`** (e.g., **`220 – 240 KIAS`**, staying below $V_{MO} = 250\text{ KIAS}$).
5. The Autothrottle (`Auto T` + `MCT`) smoothly reduces thrust to hold the target airspeed down the descent slope.
6. **Descent in `FLC`**:
7. If **`FLC`** is selected with a lower altitude, the autothrottle gently brings the engine to flight idle (`12%` TLA) while the autopilot pitches the nose down to track the dialed airspeed. Once level at the new altitude (`ALT` capture), the autothrottle automatically spools up to restore and hold the target speed.

Phase 5: Approach, Landing & Go-Around

1. **Deceleration & Aircraft Configuration**:
2. On initial approach, dial **`SPEED SET`** (in **`KTS`** mode) to **`180 KIAS`**: the autothrottle reduces thrust and stabilizes at `180 KIAS`.
3. Below **`210 KIAS`** (V_{LO}), extend landing gear (**Gear DOWN**).
4. Below **`190 KIAS`** ($V_{FE\ 50\%}$), extend **Flaps 50%** and dial **`135 – 140 KIAS`** on

the speed bug.

5. On final approach, below **`150 KIAS`** ($V_{FE\ 100\%}$), extend **Flaps 100%** and reduce the speed selector to final approach reference speed (**`85 - 90 KIAS`**; stall speed in landing configuration $V_{SO} = 67\text{ KIAS}$).
6. **Landing (Autothrottle Disengagement)**:
7. At approximately `200 - 300 ft AGL` (or prior to the landing flare), tap the **`Auto T`** button to **disengage the Autothrottle** (all `Auto T` and `MCT` annunciations extinguish). Manually retard your hardware throttle lever to **Flight Idle** for touchdown.
8. **Go-Around (`Missed Approach`)**:
9. If a missed approach is required, simply press **`TOGA`**:

    - The autothrottle advances smoothly to **100% Takeoff Thrust**.
    - Flight Director commands go-around pitch attitude (`+8°`) and sets target speed to **`160 KIAS`**.
    - Immediately retract flaps to **50%**.
    - With positive climb rate, retract the landing gear (**Gear UP**).
    - Above **`115 KIAS`**, retract flaps to **0% (UP)** and tap **`MCT`** on the GTC to resume standard climb.

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3. Cockpit Amenities & Pop-ups

- **G1000 PFD & MFD 2D Pop-ups**:
- Clicking anywhere on the active glass screen of either the pilot PFD or copilot MFD instantly pops up a resizable, high-resolution 2D window (`sim/GPS/g1000n1_popup` and `sim/GPS/g1000n3_popup`). Clicking the window again closes it.
- **GTC Overlap-Free Layout**:
- All secondary pages (Auto Throttle, Weight & Balance, and Systems) have their **Back** and **Home** navigation buttons placed on the dedicated right margin (`POS 460.0`), preventing any overlap with text boxes or system toggles.

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4. SimBrief Flight Planning Airframe

A dedicated, pre-calibrated SimBrief airframe profile matching the native XP12 Cirrus Vision Jet SF50 G2 specifications (FL310 ceiling, Williams FJ33-5A fuel flow, MTOW 6,000 lbs, accurate weights and PBN avionics) is available for 1-click import into your SimBrief fleet: * 🌐 **Click here to import the official SimBrief Airframe Profile

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5. Electronic Flight Bag (EFB), AviTab & Live Dispatch

3D Cockpit Tablet Mount

- The SF50 G2 includes a dedicated 3D tablet (EFB) attached to the left window mount (`xPad_arm.obj`).
- **Mount / Stow Interaction**: Simply click anywhere on the mount arm ball clamp or the tablet bezel in the 3D cockpit (or trigger command `laminar/sf50/tablet/toggle`) to mount or stow the tablet.

Navigraph Charts, Moving Map & SimBrief (AviTab Integration)

- Built-in native support for the free **AviTab** plugin via pre-configured `AviTab.json`:
- **Navigraph Integration**: Login with your Navigraph subscription to access geo-referenced approach plates, SIDs, STARs, and interactive airport taxi diagrams with your aircraft position.
- **Moving Map**: High-resolution GPS moving map showing terrain and flight path.
- **SimBrief OFP Viewer**: View the operational flight plan directly inside the cockpit.

Live Payload & Fuel Dispatch Manager

- The custom XLua dispatch module (`sf50_dispatch.lua`) allows real-time fueling and passenger management without pausing or opening X-Plane menus:
- **Passenger Loading**: Configure between 1 and 5 occupants (Pilot + up to 4 passengers @ 82 kg / 180 lbs each).
- **Cargo / Baggage**: Adjust luggage from 0 to 136 kg (300 lbs).
- **Quick Fuel Presets**: Instant loading for `MIN FUEL` (60 gal / 227 kg), `TABS` (140 gal /

429 kg), and `FULL TANKS` (296 gal / 907 kg), or fine adjustments in ±10 gal increments.

- **SimBrief 1-Click Load**: Command `laminar/sf50/dispatch/load_simbrief` reads your flight plan and automatically sets exact block fuel and passenger weights.
- **Instant Physics Sync**: Physical weight (`m_fixed`, `m_fuel1`, `m_fuel2`) and center-of-gravity update in real time with automated MTOW (6,000 lbs) safety status monitoring.